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Purpose or Objective

Ependymomas represent the third cause of pediatric brain tumors. Despite initial treatment including maximal surgery followed by local radiotherapy, half of the patients experience recurrence whose management is not standardized. Our study aimed at retrospectively evaluate the treatment of recurrence.

Materials and Methods

The PEPPI database gathers data of children and adolescents treated for a newly diagnosed ependymoma between 2000 and 2013, in the reference centers of SFCE. We assessed overall survival (OS) and progression free survival (PFS) after a first relapse taking into account the treatments received. Landmark analyses at 4 months after relapse, to take into account the heterogeneity of the duration of each therapeutic sequences, and cox proportional hazard models were performed.

Results

Among 202 patients included, 100 patients relapsed and 98 were analysed, 67 had a local recurrence and 31 a distant recurrence.

Median time since the initial surgery was 21,1 months. Median age at relapse was 7 years. Median follow-up was 77.8 months.

Seventy-nine patients received surgery, 22 received chemotherapy and 52 received reirradiation. Among them, radiation was stereotactic for 29 patients and normofractionated for 23 patients (among these 12 received craniospinal irradiation). Median BED was statistically lower in the stereotactic cohort (43,2 Gy (35,5 - 77,0) versus 61,6 (39,0 - 72,0), $p < 0,001$).

OS and PFS at 5 years were respectively 43,14% and 16,19%.

In multivariable analysis, chemotherapy was detrimental in both PFS (Cox model: HR 2,27, (IC95% 1,21 -4,26), $p = 0,011$. Landmark at 4 months : HR 2,40, (IC95% 1,11 -5,19), $p = 0,026$) and OS (Cox model: HR 2,28, (IC95% 1,09-4,78), $p = 0,030$. Landmark at 4 months : HR 2,95, (IC95% 1,24 -6,98), $p = 0,014$) whereas stereotactic radiation, with landmark model, is associated with better OS (HR 0,37 (IC95% 0,15 - 0,94), $p = 0,036$) and PFS (HR 0,39 (IC95% 0,19 - 0,78), $p = 0,008$).

Surgery was associated with better outcome only in univariable analysis. Patients with local relapse had better outcome with surgery and radiotherapy. A short (< 24 months) delay between the initial surgery and relapse was a major unfavourable prognostic factor.

Conclusion

The prognostic of patients after relapse is poor. But, in the PEPPI study, treatment including stereotactic radiotherapy and surgery improves survival while chemotherapy shows no benefit. Patients with an early relapse have an unfavourable prognosis.

PH-0327 Tumour control and visual outcomes after proton therapy for uveal melanoma

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Purpose or Objective

To assess oncological and visual outcomes after proton therapy for choroidal melanoma and to determine factors for loss of visual acuity.

Materials and Methods

This is a retrospective study of Dutch patients with large and/or juxtapapillary uveal melanomas treated with proton therapy abroad (mainly in Switzerland) from 1987-2019. All patients were treated with 60.0 CGE in 4 fractions using EYEPLAN planning software after placement of localization clips. Actuarial estimates of oncological outcomes, secondary enucleation and visual acuity were calculated according to Kaplan-Meier's methodology. Local recurrence included vital tumor rests as defined by histology in eyes that were enucleated for toxicity. Risk factors for visual acuity <0.2 were identified using Cox' proportional hazards models. Visual acuity before and at 1-5 years after proton therapy were analyzed by hierarchical clustering to show visual acuity over time in relation to tumor location and dose parameters.

Results

There were 103 patients (104 eyes); 55 were male (53.4%), mean age was 58 years (range 24-85). Median tumor diameter and height were 18.7 (IQR 3.8) and 8.4 mm (IQR 3.2) respectively (median volume 1162.5mm³, IQR 823). Tumors were localized centrally in 10.6%, mid-peripherally in 65.4% and peripherally in 33.7%. Tumor stages were T1 in 2.9%, T2 in 8.7%, T3 in 26.0% and T4 in 62.5%. Median follow-up was 7.0 years (SE 1.0). Disease-specific survival was 72.6% (SE 5.2) at 5-years. Distant metastasis, local recurrence and enucleation had occurred at 5 years in respectively 29.8% (SE 5.4%), 14.9% (SE 3.8) and 20.1% (SE 4.3%). Visual